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Configurable integrated power delivery module with adaptive power sharing

Abstract

A multi-port charger includes two or more integrated power delivery modules electrically coupled to an AC-to-DC power converter. Each of the integrated power delivery modules includes a module controller in signal communication with a digital communication bus, a USB-PD controller, a switch-mode DC-to-DC power converter which is configured to provide an adjustable output voltage to a sink device via a USB voltage bus, a first analog-to-digital converter (ADC) circuit in signal communication with the USB-PD controller and the USB voltage bus to generate a digital representation of the output voltage, and a second ADC circuit in signal communication with the USB-PD controller and the USB voltage bus to provide a digital representation of an output current provided by the switch-mode DC-to-DC power converter to the sink device.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/337,928, filed Jun. 20, 2023, which claims priority to U.S. Provisional Patent Application No. 63/368,448, filed Jul. 14, 2022, all of which is incorporated herein by reference in its entirety for all purposes.

BACKGROUND

(1) Many mobile devices, such as cell phones, laptop computers, tablet computers, and similar, are shipped from their respective manufacturer with a single charger. Such chargers typically receive AC power and produce a DC power output on a single port in accordance with the USB-PD (USB Power Delivery) industry standard. However, some consumers may wish to have a multi-port charger that can supply a DC power output to more than one upward-facing port device (“sink device”) at a time. For example, such consumers may desire a multi-port charger that is operable to charge their cell phone and wireless headphones simultaneously. However, multi-port chargers often must adhere to strict maximum power output limits (e.g., 45 W) as a function of charger spatial volume to limit a maximum temperature thereof. Therefore, even if a manufacturer includes two entirely separate charger circuits within a single package to implement a multi-port charger, as is conventionally done, each of those charger circuits must be power limited to provide only a fixed fraction of the total maximum power output limit. For example, two charger circuits in a single package would be limited by the manufacturer to only provide half of a total maximum power output limit. As a result, sink devices plugged into such multi-port chargers will not charge as rapidly as compared to plugging each sink device into a discrete charger.

(2) Additionally, existing multi-port chargers solutions that address some of the problems outlined herein conventionally do not provide substantial flexibility with regard to configurability or power-efficient modes of operation.

SUMMARY

(3) In some embodiments, a multi-port charger includes an AC-to-DC power converter that receives an AC input voltage and generates a DC output voltage therefrom, and two or more integrated power delivery modules electrically coupled to the AC-to-DC power converter and in signal communication with a digital communication bus. Each of the integrated power delivery modules includes a module controller in signal communication with the digital communication bus, a USB-PD controller (“PD controller”) in signal communication with the module controller and which is configured to be connected to a sink device adhering to a USB standard, a switch-mode DC-to-DC power converter in signal communication with the module controller and the PD controller and which is configured to provide an adjustable output voltage to the sink device via a USB voltage bus, a first analog-to-digital converter (ADC) circuit in signal communication with the PD controller and the USB voltage bus to generate a digital representation of the output voltage provided by the switch-mode DC-to-DC power converter to the sink device, and a second ADC circuit in signal communication with the PD controller and the USB voltage bus to provide a digital representation of an output current provided by the switch-mode DC-to-DC power converter to the sink device.

(4) In some embodiments, an integrated power delivery module includes a module controller in signal communication with a digital communication bus, a USB-PD controller (“PD controller”) in signal communication with the module controller and which is configured to be connected to a sink device adhering to a USB standard, a switch-mode DC-to-DC power converter in signal communication with the module controller and the PD controller and which is configured to

receive an input voltage and to provide an adjustable output voltage to the sink device via a USB voltage bus, a first analog-to-digital converter (ADC) circuit in signal communication with the PD controller and the USB voltage bus to generate a digital representation of the output voltage provided by the switch-mode DC-to-DC power converter to the sink device, and a second ADC circuit in signal communication with the PD controller and the USB voltage bus to provide a digital representation of an output current provided by the switch-mode DC-to-DC power converter to the sink device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a simplified schematic of a prior-art multi-port charger.
- (2) FIG. 2 is a simplified schematic of a multi-port charger with adaptive power sharing, in accordance with some embodiments.
- (3) FIG. 3 is a simplified schematic of a circuit that includes a configurable integrated power delivery module of the multi-port charger shown in FIG. 2, in accordance with some embodiments.
- (4) FIG. 4A is a simplified schematic of a circuit providing select details of the configurable integrated power delivery module included in the circuit shown in FIG. 3, in accordance with some embodiments.
- (5) FIG. 4B is a simplified schematic of a circuit providing select details of the DC-to-DC power converter circuit included in the configurable integrated power delivery module shown in FIG. 4A, in accordance with some embodiments.
- (6) FIG. 5 through FIG. 17 provide a simplified example process for adaptive power-sharing using the multi-port charger shown in FIG. 2, in accordance with some embodiments.

DETAILED DESCRIPTION

- (7) Some consumers desire a multi-port charger that is operable to charge multiple sink devices simultaneously. However, many conventional multi-port charger implementations have limited (or zero) flexibility for making power contracts with sink devices based upon sink power requests and actual power consumption of the sink devices with respect to the available power of the multi-port charger. Therefore, even if a manufacturer includes two entirely separate charger circuits within a single package to implement a multi-port charger, which is conventionally done, each of those charger circuits must be power limited to provide only a fraction of the total maximum power output limit. Additionally, some conventional solutions that distribute power between multiple ports do so in a fixed, non-configurable, and/or coarse manner, which limits flexibility and power efficiency as compared to the techniques disclosed herein.
- (8) Disclosed herein is an integrated power delivery module that advantageously communicates with one or more other integrated power delivery modules of a multi-port charger. The integrated power deliver modules adaptively and continually control how much power is delivered to each port of the multi-port charger with high-granularity as power demands of the connected sink devices change over time based on calculated available power. The integrated power delivery modules adaptively and continually control how much power is delivered to each port of the multi-port charger in response to temperature changes, priority, battery charge levels, and other status events of the connected sink devices. The integrated power delivery modules are named as such because a USB-PD controller is integrated into the same package as a DC-to-DC power converter. They are operable to configure the DC-to-DC power converter therein into a low-power mode when no sink device is connected to the port associated with that integrated power delivery module.
- (9) FIG. 1 is a simplified schematic of a prior-art multi-port charger **101** that is connected to sink devices **112a-b** (e.g., cell phones). The multi-port charger **101** includes two independent charger circuits. A first one of the independent charger circuits of the multi-port charger **101** includes an

AC-to-DC (“AC/DC”) power converter circuit **102a**, an AC-to-DC power converter control circuit **104a**, a DC-to-DC (“DC/DC”) power converter circuit **106a**, and a USB-PD control circuit (“PD control”) **108a**, connected as shown. A second one of the independent charger circuits of the multi-port charger **101** includes an AC-to-DC power converter circuit **102b**, an AC-to-DC power converter control circuit **104b**, a DC-to-DC power converter circuit **106b**, and a USB-PD control circuit (“PD control”) **108b**, connected as shown. Each of the independent charger circuits of the multi-port charger **101** receives an AC voltage AC.sub.in and produces a respective DC voltage Vin.sup.a and Vin.sup.b therefrom. The DC-to-DC power converter circuits **106a** and **106b** respectively receive the DC voltages Vin.sup.a and Vin.sup.b and produce a respective USB bus voltage VBUS.sup.a and VBUS.sup.b therefrom. The USB-PD control circuit **108a** produces signals CC1.sup.a, CC2.sup.a, D+.sup.a, and D-.sup.a, in accordance with the USB-PD standard. Similarly, the USB-PD control circuit **108b** produces signals CC1.sup.b, CC2.sup.b, D+.sup.b, and D-.sup.b, in accordance with the USB-PD standard. As mentioned above, to comply with the maximum power limit of the multi-port charger **101**, each of the independent charger circuits therein is conventionally power limited such that each only provides a fixed percentage of the maximum power limit.

(10) FIG. 2 is a simplified schematic of a multi-port charger **201** with configurable and adaptive power-sharing that is connected to sink devices **212p-q** (e.g., cell phones), in accordance with some embodiments. The multi-port charger **201** includes a single AC-to-DC (“AC/DC”) power converter **202**, a single AC-to-DC control circuit **204**, and multiple integrated power delivery modules (“Integrated PD Module”, or “IPD Module”) **220p-q**. Each port, p through q, of the multi-port charger **201** has a corresponding respective integrated PD module that includes a respective USB PD controller circuit, a respective module controller circuit, and a respective switch-mode DC-to-DC power converter, as described below.

(11) The sink device **212p** is electrically and communicatively coupled to port p of the multi-port charger **201** by the integrated PD module **220p**. Similarly, the sink device **212q** is electrically and communicatively coupled to port q of the multi-port charger **201** by the integrated PD module **220q**. Some elements of the multi-port charger **201** have been omitted to simplify the description thereof but would be understood by one of ordinary skill in the art to be present.

(12) As shown, the single AC-to-DC power converter **202** receives an AC input voltage AC.sub.in and produces a shared DC voltage rail Vin therefrom. Each of the integrated PD modules **220p** through **220q** receives the DC voltage Vin and respectively produces a USB bus voltage VBUS.sup.p through VBUS.sup.q therefrom using an integrated switch-mode DC-to-DC power converter circuit. The integrated PD module **220p** produces signals CC1.sup.p, CC2.sup.p, D+.sup.p, and D-.sup.p at port p, in accordance with the USB-PD standard. Similarly, the integrated PD module **220q** produces signals CC1.sup.q, CC2.sup.q, D+.sup.q, and D-.sup.q, in accordance with the USB-PD standard. As shown by a line therebetween, the integrated PD module **220p** and the integrated PD module **220q** are advantageous communicatively coupled to each other via a digital communication bus “Comm (SDA/SCL)” (e.g., a serial or parallel data bus, such as a data bus that adheres to the I2C or SPI standard). Because the integrated PD module **220p** and the integrated PD module **220q** are communicatively coupled to each other, the integrated PD modules **220p-q** are operable to communicate with one another to continually and adaptively update the amount of power delivered to each port of the multi-port charger **201** with fine control.

(13) By comparison, some conventional distributed solutions either communicate an available amount of power using a shared analog bus having fixed resistor values or may communicate with a single power delivery coordination circuit. Such conventional solutions lack the flexibility, configurability, and granularity of control as compared to the integrated PD modules disclosed herein. For example, some conventional multi-port chargers may be operable to distribute a fixed amount of power between multiple sink devices, but may not be able to adjust how much power a first sink device is receiving based on changing device priorities and/or the status of two or more

second sink devices connected to the multi-port charger.

(14) FIG. 3 is a simplified schematic of a circuit **300** that includes an integrated PD module **320** that is similar to the integrated PD modules **220p-q** of the multi-port charger **201** shown in FIG. 2, in accordance with some embodiments. The circuit **300** generally includes the integrated PD module **320**, resistors R1-R5, capacitors C1-C7, a switch M1, an inductor L1, and thermistor R_{therm}. Also shown are signal nodes of the circuit **300**, which include signal and voltage nodes designated as VIN, AVIN, VREG, AVREG, VDD1P5, EN, SDA, SCL, ALERT, PGOOD, RCO, RC1, RC2/NTC, CC1, CC2, DP, DM, VBUS, DISCSW, ISNS, VOUTSNS, BOOT, PH, PGND, AGND, and NTC/GPIO. Description of some of the nodes shown in FIG. 3 are omitted herein for brevity.

(15) The nodes designated CC1 and CC2 are part of a configuration and communication channel for USB-PD communication with a sink device, the nodes designated DP and DM comprise a communication channel for USB-PD fast charging communication with a sink device, and the node designated VBUS of a USB voltage bus provides an output voltage to a sink device as well as serving as voltage sense line, as shown in FIG. 4A. The node designated as PH is a phase switch node for a switch-mode DC-to-DC power converter that is advantageously internal to the circuit **300** as described below. The inductor L1 and the capacitor C7 provide an output filter stage of the internal switch-mode DC-to-DC power converter.

(16) The nodes RCO, RC1, and RC2/NTC are resistor configuration nodes used to set operational parameters of the integrated PD module **320**, which are described in more detail below. The node NTC/GPIO is operable to be connected to a temperature sensing circuit (e.g., a thermistor) to provide a temperature measurement of, or near to, the integrated PD module **320**.

(17) Also shown are signals designated as CC1 Signals, CC2 Signals, D+ Signals, D- Signals, a VIN Voltage, a VBUS Voltage, and a PH Signal. Some elements and signals of the circuit **300** have been omitted to simplify the description thereof but would be understood by one of ordinary skill in the art to be present. Details of the integrated PD module **320** are described below.

(18) FIG. 4A is a simplified schematic of a portion of the integrated power delivery module **320** shown in FIG. 3, in accordance with some embodiments. As shown, the integrated power delivery (PD) module **320** generally includes a module controller **402** (e.g., implemented using a microcontroller, a microprocessor, an FPGA, and/or an ASIC), a switch-mode DC-to-DC (“DC/DC”) power converter **404** (e.g., implemented as a switched buck-mode converter utilizing external components L1 and C7), a USB-PD Controller (“PD controller”) **406** to provide an adjustable DC output voltage, one or more volatile and/or non-volatile memory blocks **408** which may be part of the module controller **402** and/or the PD controller **406** or may be one or more separate modules, multiple analog-to-digital (“ADC”) converter circuits **410**, programmatically controlled termination (Pull-up/Pull-down) resistors (e.g. R_p) (“PU/PD Resistors”) **412**, and a signal multiplexing circuit (“MUX”) **414**, connected as shown. Some elements of the integrated PD module **320** have been omitted to simplify the description thereof but would be understood by one of ordinary skill in the art to be present. Also shown are control signals CTRL_{sup}.1-4, a configuration signal CFG(n), the previously introduced USB protocol signals designated as CC1 Signals, CC2 Signals, VBUS Voltage, D+ Signals, D- Signals, analog current sense ISNS signals, analog voltage sense VOUTSNS signals, an analog USB bus voltage sense signal VBUS Voltage, analog temperature measurement sense NTC/GPIO Signals, a digital representation of the analog current sense signal ISNS(n), a digital representation of the analog voltage sense signal VOUTSNS(n), a digital representation of the USB bus voltage VBUSV(n), a digital representation of an analog temperature measurement signal NTC(n), the phase node signal PH from the DC-to-DC power converter circuit **404**, and communication signals SDA/SCL of a digital communication bus designated as Comm. Also shown are previously introduced nodes SDA, SCL, PH, VBUS, CC1, CC2, DP, DM, ISNS, VOUTSNS, and NTC/GPIO.

(19) The memory block **408** is advantageously operable to store programmable (e.g., from an

external interface, not shown) configurations of the integrated PD module **320**, such as a maximum or total allowable power that can be delivered by the integrated PD module **320**. The module controller **402** is operable to retrieve the programmable configurations from the memory block **408** via the configuration signal CFG(n) and to control the PD controller **406** and the DC-to-DC power converter circuit **404** in accordance with the retrieved programmable configurations. The module controller **402** is also operable to communicate with respective module controllers of other integrated power delivery modules of a multi-port charger, e.g., using communication signals SDA/SCL over the digital communication bus Comm, to continually and adaptively control how much power may be provided to a connected sink device by each integrated PD module **320**. The module controller **402** advantageously enables each integrated PD module **320** of a multi-port charger to be configured to precisely deliver a desired amount of power to a connected sink device.

(20) The ADC circuit **410** includes multiple ADC circuits, or one or more multiplexed ADC circuits, and is operable to receive analog signals and to create digital representations thereof. As shown, the ADC circuit **410** receives an analog current sense signal ISNS, an analog output voltage sense signal VOUTSNS, an analog VBUS voltage, and an analog temperature sense signal NTC/GPIO. The ADC circuit **410** uses the aforementioned received analog signals to create respective digital representations ISNS(n), VOUTSNS(n), VBUSV(n), and NTC(n).

(21) The PD controller **406** is operable to use the respective digital representations for making USB control and policy decisions and is further operable to transmit the respective digital representations to the module controller **402**. The module controller is operable to use the digital representations of the current sense signal ISNS(n) and the digital representation of the VBUS Voltage VBUSV(n) to calculate (e.g., by multiplying the values thereof) an actual amount of power that is being provided by the DC-to-DC power converter **404** to a sink device. Each PD controller **406** advantageously receives digital signals ISNS(s) and VBUSV(n) which are representative of sensed current and voltage, respectively, to manage power delivery to the sink device by controlling the DC-to-DC power converter **404** and/or the power contract established with the sink device.

(22) The PD controller **406** includes modules (not shown) that implement the USB Power Delivery (PD) protocol to exchange commands and messages to negotiate and establish power contracts between each integrated PD module **320** and a sink device connected thereto, such as a mobile phone or notebook. The PD controller **406** communicates with the module controller **402** to advantageously coordinate and negotiate power distribution between other respective PD controllers **406**. As shown in FIG. 4A, each PD controller **406** is operable to communicate with a sink device via the CC1, CC2, DP, and DM nodes.

(23) Some sink devices require constant current and some sink devices require constant voltage. How much voltage and current is needed by a particular connected sink device is communicated by the PD controller **406** to the module controller **402**, and the module controller **402** determines if the multi-port charger has enough available power remaining to deliver for that request. The module controller **402** communicates (e.g., periodically such as every 5 second, 10 seconds, 20 seconds, or another appropriate amount of time, or in response to an event) with module controllers of the other integrated PD controllers to determine the current status of total power already delivered and to calculate how much additional charger power remains available. The module controller **402** is further operable to continuously and optimally re-distribute power contracts to already connected sink devices of the multi-port charger based on changing priorities or status events of the connected sink devices. The module controller **402** and the PD controller **406** thereby advantageously manage power sharing and power allocation and power re-balancing for a multi-port charger to ensure that the total power delivered to all ports will not exceed the total power capacity of the multi-port charger.

(24) The PD controller **406** is operable to generate an output voltage setpoint of the DC-to-DC power converter **404**, using the control signal CTRL.sup.3, such that the power provided to a sink

device connected to the integrated PD module **320** is advantageously only slightly above, within some margin, to what the sink device requires, thereby increasing energy efficiency as compared to conventional solutions.

(25) The PD controller **406** and module controller **402** of each of the integrated PD modules **320** are advantageously aware of all port statuses of the multi-port charger **201**. Therefore, in some embodiments, the PD controller **406** and/or the module controller **402** are aware if no ports of the multi-port charger **201** are connected to sink devices and are operable to place each DC-to-DC power converter into a low-power standby mode.

(26) In some embodiments, the module controller **402** manages power balancing to each port of the multi-port charger **201** in granular steps, such as 2 W per 10 seconds, and power balancing is advantageously performed without the need for port resets and/or re-established handshakes.

(27) The module controller **402** also advantageously communicates to the DC-to-DC power converter **404** via control signal CTRL.sup.2 when one or more sink devices of the multi-port charger are not USB-PD compliant but is instead a normal battery charger load. In such instances, the DC-to-DC power converter **404** is set by the module controller **402** to a fixed power initially and then updated periodically. For example, the DC-to-DC power converter **404** may increase the power delivered to a load by 2 W every 10 seconds if power is still available.

(28) The PD controller **406** and/or the module controller **402** are advantageously operable to use the digital representations ISNS(n) and VBUSV(n) to continually and adaptively determine (e.g., by multiplying the values thereof) and control an actual amount power that is delivered by the integrated PD module **320** by communicating with other integrated power delivery modules of a multi-port charger circuit. By comparison, some conventional solutions may use a shared analog power line to determine how much power is being delivered by the combined conventional power delivery modules. As disclosed herein, by calculating, using the module controller **402**, how much power is being delivered by a respective integrated PD module **320**, the module controller has greater flexibility in being able to change operating modes based on user configurations and preferences. For example, based on which type of sink device is connected to a particular integrated PD module **320**, the module controller **420** thereof may adaptively control maximum and minimum power delivery settings.

(29) FIG. **4B** is a simplified schematic of a circuit providing select details of the DC-to-DC power converter circuit **404** included in the circuit shown in FIG. **4A**, in accordance with some embodiments. As shown, the DC-to-DC power converter circuit **404** includes a buck converter controller **422**, a ramp-generator circuit **424**, an on-time generator circuit **426**, a signal summation circuit **428**, a reference voltage generator circuit **430**, a high-side (“HS”) gate driver circuit **440** for a high-side switch MH, a low-side (“LS”) gate driver circuit **442** for a low-side switch ML, a low-side FET current sense circuit **444**, and a fault management circuit **446**, connected as shown. Also shown are the previously introduced nodes RCO, VIN, PH, PGND, and VOUT, as well as signals CTRL.sup.3, OC, PH Signal, VREF, VIN, VOUT, and “fsw, t.sup.ss Select”.

(30) The reference voltage generator **430** is operable to receive the control signal CTR.sup.3 from the PD controller circuit **406** shown in FIG. **4A** to generate a reference voltage level VREF for configuring a desired VBUS output voltage based on a negotiated amount of power, voltage, and/or current to be delivered to a sink device connected thereto. The PD controller is additionally operable to adjust the reference voltage level VREF in response to the digital representation ISNS(s) of a sensed output current generated by the DC-to-DC power converter circuit **404**, the digital representation VOUTSNS(n) of a sensed output voltage generated by the DC-to-DC power converter circuit **404**, and/or the digital representation VBUSV(n) of a sensed USB bus voltage generated by the DC-to-DC power converter circuit **404**. The fault management circuit is operable to receive an overcurrent alert signal OC, the digital representation of the output voltage VOUTSNS(n), as well as other signals, such as an indication that the input voltage is undervoltage (not shown) to halt or adjust the operation of the DC-to-DC power converter **404**.

(31) The buck converter controller circuit **422** is operable to receive the control signal CTRL.sup.2 from the module controller **402** and/or the control signal CTRL.sup.3 from the PD controller circuit **406** shown in FIG. 4A to change operating parameters and other configuration settings. For example, if the PD controller **406** determines that no sink device is connected to the integrated PD module **320**, the DC-to-DC power converter circuit **404** may be placed in a low-power mode. In low-power mode, some or all switching signals of the DC-to-DC power converter **404** (e.g., of the ramp-generator circuit **424**, and of the switches MH and ML) may be disabled to conserve power. Additionally, the buck converter controller circuit **422** is operable to receive configuration settings from node RCO that include a maximum switching frequency f_s , and soft start time t_{ss} , as well as configuration settings from the module controller **402** and/or the PD controller **406**.

(32) As compared to conventional solutions, the DC-to-DC power converter **404** is configurable on a per-port basis of a multi-port converter and the configuration settings may be updated on an ongoing basis as operational conditions change.

(33) FIG. 5 provides a portion of a simplified example process **500** for adaptive power-sharing using the multi-port charger **201** shown in FIG. 2, in accordance with some embodiments. The particular steps, the order of steps, and the combination of steps are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(34) At step **501**, a maximum current for each port (i.e., p through q) of the multi-port charger **201** is set by the integrated PD modules 220_{p-q} to be 1.5 A if a Type-C standard is used for those ports. In some embodiments, one of the integrated PD modules 220_{p-q} acts as a master controller, and each of the remaining integrated PD modules 220_{p-q} acts as a respective slave controller. Thus, in such embodiments, the master integrated PD module commands the slave integrated PD modules to perform each of the steps described herein. By comparison, some conventional multi-port chargers rely on a single policy controller circuit that provides power delivery settings to each power delivery module thereof.

(35) At step **502**, a maximum available power $P_{sub.available}$ that remains to be distributed to all ports of the multi-port charger **201** is set to a total allowable power P_{total} for the multi-port charger **201** (e.g., as specified by programmable configurations stored at the memory block **408** shown in FIG. 4A). For example, if the maximum available power $P_{sub.available}$ is equal to 15 W, 15 W may be distributed between the integrated PD modules 220_{p-q} of the multi-port charger **201**. In some embodiments, such distribution may be based on a fixed or changing priority of the ports and/or the connected sink devices. The priority may advantageously be configured at the time of manufacturing (e.g., based on a configuration resistor), may be programmatically configured during operation of the multi-port charger **201** (e.g., a programmed configuration setting may assign a particular port a greater priority), or may be based on a device identifier of a connected sink device (e.g., a user may configure the multi-port charger such that their phone always has a higher priority for charging as compared to a priority assigned to charging wireless headphones). Additionally, port priority may be updated automatically during the operation of the multi-port charger based on the status of connected sink devices as well as other factors, such as changing battery charge levels of respective batteries of the sink devices, port temperatures, a powered status of the connected sink devices (e.g., a sink device that is powered on may receive a higher power allocation as compared to a sink device that is off and is merely being recharged), or other status.

(36) For example, port p may be allocated a power output of $p_{sub.alloc.sup.p}=15$ W, and port q may be allocated a power output of $p_{sub.alloc.sup.q}=0$ W. Or, port p may be allocated a power output of $p_{sub.alloc.sup.p}=10$ W, and port q may be allocated a power output of $p_{sub.allo.sup.q}=5$ W. Or, port p may be allocated a power output of $p_{sub.alloc.sup.p}=7.5$ W, and port q may be allocated a power output of $p_{sub.alloc.sup.q}=7.5$ W, and so on. This adaptive allocation occurs continually (e.g., every 5 s, 10 s, 15 s, or at another appropriate update rate) as the power requirements, status, and/or states of sink devices connected to the multi-port charger **201** change.

For example, if two sink devices having completely drained batteries are connected to the multi-port charger **201**, a first sink device connected to the master integrated PD module will initially receive a maximum allocated power and a second sink device connected to a slave integrated PD module will initially receive a minimum allocated power. As the first sink device charges, the power required by that sink device will decrease. As the power required by the first sink device decreases, the integrated PD modules of the multi-port charger **201** adaptively increase the power delivered to the second sink device and decrease the power delivered to the first sink device. In some embodiments, P.sub.available is stored at a master integrated PD module of the multi-port charger **201**. In other embodiments, P.sub.available is stored at each integrated PD module of the multi-port charger **201**.

(37) At step **503**, the total power allocated to each port p-q of the multi-port charger **201** is initialized to 0 W. At step **504**, USB event detection is enabled at each port p-q of the multi-port charger **201**. At step **506**, each integrated PD module of the multi-port charger **201** waits for event detection at the port that corresponds to that integrated PD module. Flow may continue to step **508** or step **1202** (shown in FIG. **12**) based on which event was determined to have occurred. The series of steps **508** through **516** may be performed in series or in parallel with the series of steps **1202** through **1214** shown in FIG. **12**.

(38) Upon detecting a USB event at one or more ports at step **506**, the steps that follow are described with reference to USB events detected specifically at port p of the multi-port charger **201** using the integrated PD module **220p** for simplicity. However, similar, or the same steps are followed for USB events detected at any of the other ports p-q of the multi-port charger **201**.

(39) At step **508**, if a Type-C connection was detected at port p, flow of the process **500** continues to step **602** shown in FIG. **6**. Otherwise, flow continues to step **510**. At step **510**, if a USB Standard BC1.2 (USB Battery Charging version 1.2) connection was detected at port p, flow of the process continues to step **702** shown in FIG. **7**. Otherwise, flow continues to step **512**. At step **512**, if a quick charge sink connection was detected at port p, flow of the process continues to step **902** shown in FIG. **9**. Otherwise, flow continues to step **514**. At step **514**, if power contract negotiation is complete, in accordance with the USB-PD standard, flow continues to step **1002** shown in FIG. **10**. Otherwise, flow continues to step **516**. At step **516**, if a sink device disconnection was detected at port p, flow continues to step **1102** shown in FIG. **11**. Otherwise, flow returns to step **506**.

(40) FIG. **6** provides a portion of a simplified example process **600** for adaptive power-sharing using the multi-port charger **201** shown in FIG. **2**, in accordance with some embodiments. The particular steps, the order of steps, and the combination of steps are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(41) Step **602** of the process **600** continues from step **508** shown in FIG. **5** and is performed in response to a determination at step **508** that a Type-C connection was detected at port p. At step **602**, if it is determined by negotiating between the module controllers **402** of the integrated PD modules **220p-q** of the multi-port charger **201** using the digital communication bus Comm that the maximum available power P.sub.available that remains to be distributed between the ports of the multi-port charger **201** is greater than or equal to a target amount of power (e.g., 15 W), flow continues to step **604**. Otherwise, flow continues to step **802** shown in FIG. **8**.

(42) At step **604**, a maximum current for port p is set to 3 A by configuring the programmatically controlled termination resistors **412** and signal multiplexing circuit **414**, using the PD controller **406** via the control signal CTRL.sup.4, to values indicative of R_p 3.0 (e.g., about 10 k Ohms), per the USB-PD standard, and updating a setting of the DC-to-DC power converter **404** if needed). At step **606**, the target allocated power P.sub.alloc.sup.p for port p is set to 15 W. As such, port p of the multi-port charger **201** will deliver up to, but no more than, 15 W of power to a sink device connected to port p of the multi-port charger **201** and a setting of the DC-to-DC power converter **404** is updated accordingly if needed. At step **608**, because 15 W of power has been allocated to

port p, the maximum available power $P_{sub.available}$ of the multi-port charger **201** that remains to be distributed between the ports thereof is reduced by 15 W. The adjustment in maximum available power $P_{sub.available}$ is communicated to one or more other integrated PD modules **220p-q** by the module controller **402** via the digital communication bus Comm. At step **610**, the USB-PD contract $P_{sub.contract.sup.p}$ for port p is set to 15 W, in accordance with the USB-PD standard. At step **612**, USB-PD contract negotiation for port p is initiated by the integrated PD module in accordance with the USB-PD standard. Flow of the process then returns to step **506** shown in FIG. 5.

(43) FIG. 7 provides a portion of a simplified example process **700** for adaptive power-sharing using the multi-port charger **201** shown in FIG. 2, in accordance with some embodiments. The particular steps, the order of steps, and the combination of steps are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(44) Step **702** of the process **700** continues from step **510** shown in FIG. 5 and is conducted in response to a determination at step **510** that a BC1.2 connection was detected at port p. At step **702**, if it is determined that the power $P_{sub.alloc.sup.p}$ allocated to port p of the multi-port charger **201** (i.e., via communication between the integrated PD modules thereof using the digital communication bus Comm) is greater than or equal to a target amount of power (e.g., 7.5 W), flow of the process **700** continues to step **704**. Otherwise, flow continues to step **802** shown in FIG. 8. At step **704**, quick charge detection is enabled for port p of the multi-port charger **201**. Flow of the process then continues back to step **506** shown in FIG. 5.

(45) FIG. 8 provides a portion of a simplified example process **800** for adaptive power-sharing using the multi-port charger **201** shown in FIG. 2, in accordance with some embodiments. The particular steps, the order of steps, and the combination of steps are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(46) Step **802** of the process **800** continues from either step **602** shown in FIG. 6, or from step **702** shown in FIG. 7. At step **802**, if it is determined by negotiating between the module controllers **402** of the integrated PD modules **220p-q** of the multi-port charger **201** using the digital communication bus Comm that the maximum available power $P_{sub.available}$ of the multi-port charger **201** that remains to be distributed between the integrated PD modules thereof is greater than or equal to a target amount of power (e.g., 7.5 W), flow of the process **800** continues to step **804**. At step **804**, the maximum available power $P_{sub.available}$ of the multi-port charger **201** that remains to be distributed to ports thereof is reduced by 7.5 W. The adjustment in maximum available power $P_{sub.available}$ is communicated to one or more other integrated PD modules **220p-q** by the module controller **402** via the digital communication bus Comm. At step **806**, if it is determined that a USB Type-C connection was detected at port p, flow continues to step **816**. At step **816**, a maximum current for port p is programmatically set to 1.5 A by configuring the programmatically controlled termination resistors **412** and signal multiplexing circuit **414**, using the PD controller **406** via the control signal CTRL.sup.4, to values indicative of R_p 1.5 (e.g., about 22 k Ohms), per the USB-PD standard. Conventional solutions may use fixed resistor termination resistor values, designated in the USB standard as R_p , which determine a maximum current-carrying capability of a power source. As disclosed herein, the PD controller **406** is operable to adaptively adjust, by controlling the DC-to-DC power converter, how much current can be provided to a sink device by the integrated PD module.

(47) At step **818**, the target allocated power $P_{sub.alloc.sup.p}$ for port p is set to 7.5 W. At step **820**, several flags are set by the PD controller **406** for port p, including a Less Power Flag and a No PD Flag, in accordance with the USB-PD standard. These flags are asserted when the power requested by a sink device cannot be supplied by the port associated with that sink device (e.g., not enough power has been allocated to that port). The asserted flags indicate to the multi-port charger **201** that more power should be supplied to the sink device as more power becomes available. Flow then

returns to step **506** shown in FIG. 5.

(48) If it was determined at step **802** that $P_{sub.available}$ is not greater than or equal to (i.e., is less than) 7.5 W, flow of the process **800** continues to step **808** to advantageously reduce power allocated to another port of the multi-port charger **201**. At step **808**, the integrated PD modules of the multi-port charger **201** communicate between themselves using module controllers **402** thereof via the digital communication bus Comm to identify a port of the multi-port charger **201**, (e.g., port q), that currently has the maximum allocated power, e.g., $P_{sub.alloc.sup.q}$. That is, in this example, port q has the current maximum allocated power. At step **810**, the allocated power $P_{sub.alloc.sup.q}$ for port q is reduced by 7.5 W. At step **812**, the USB-PD contract $P_{sub.contract.sup.q}$ for port q is set to $P_{sub.alloc.sup.q}$. At step **814**, USB-PD contract negotiation for port q is initiated by the integrated PD module associated with port q (e.g., the integrated PD module **220 q**), in accordance with the USB-PD standard. Flow of the process **800** then continues to step **806** which was described above.

(49) If it was determined at step **806** that the connection at port p is not USB Type-C, flow continues to step **822**. At step **822**, the allocated power $P_{sub.alloc.sup.p}$ for port p is set to 7.5 W. At step **824**, quick charge detection is enabled for port p . Flow then returns to step **506** shown in FIG. 5.

(50) FIG. 9 provides a portion of a simplified example process **900** for adaptive power-sharing using the multi-port charger **201** shown in FIG. 2, in accordance with some embodiments. The particular steps, the order of steps, and the combination of steps are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(51) Step **902** of the process **900** continues from step **512** shown in FIG. 5 and is conducted in response to a determination at step **512** that a USB quick charge sink device connection was detected at port p . If it is determined by negotiating between the module controllers **402** of the integrated PD modules **220 $p-q$** of the multi-port charger **201** using the digital communication bus Comm that the maximum available power $P_{sub.available}$ of the multi-port charger **201** that remains to be distributed between the ports thereof is greater than or equal to a target amount of power (e.g., 18 W) minus the power $P_{sub.alloc.sup.p}$ currently allocated to port p , flow of the process **900** continues to step **904**. At step **904**, USB quick charge class A mode is enabled for port p , in accordance with the USB-PD standard. At step **906**, the maximum available power $P_{sub.available}$ of the multi-port charger **201** that remains to be distributed between the ports thereof is reduced by 18 W and the amount of power $P_{sub.alloc.sup.p}$ previously allocated to port p is added back to the maximum available power $P_{sub.available}$. The adjustment in maximum available power $P_{sub.available}$ is communicated to one or more other integrated PD modules **220 $p-q$** by the module controller **402** via the digital communication bus Comm. Accordingly, at step **908**, the target amount of power $P_{sub.alloc.sup.p}$ allocated to port p is updated to 18 W. Flow then returns to step **506** shown in FIG. 5.

(52) If it was determined at step **902** that the maximum available power $P_{sub.available}$ of the multi-port charger **201** that remains to be distributed to ports thereof is not greater than or equal to (i.e., is less than) 18 W minus the power $P_{sub.alloc.sup.p}$ currently allocated to port p , flow of the process **900** continues to step **910**. At step **910**, USB quick charge mode is disabled for port p , in accordance with the USB-PD standard. Additionally, at step **912**, several flags are set for port p , including a Set Less Power and No Quick Charge, in accordance with the USB-PD standard. Flow then returns to step **506** shown in FIG. 5.

(53) FIG. 10 provides a portion of a simplified example process **1000** for adaptive power-sharing using the multi-port charger **201** shown in FIG. 2, in accordance with some embodiments. The particular steps, the order of steps, and the combination of steps are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(54) Step **1002** of the process **1000** continues from step **514** shown in FIG. 5 and is conducted in response to a determination at step **514** that completion of a USB power contract negotiation, in accordance with the USB-PD standard, was detected at port p. Additionally, USB capability mismatch for the sink device at port p may have occurred. USB capability mismatch occurs when a sink device cannot satisfy its power requirements from the capabilities offered by the source (i.e., the power delivered by port p). At step **1002**, the maximum available power $P_{sub.available}$ of the multi-port charger **201** that remains to be distributed between the ports thereof is reduced by the negotiated power $P_{sub.contract}$, and the amount of power $P_{sub.alloc.sup.p}$ previously allocated to port p is added back to the maximum available power $P_{sub.available}$. The adjustment in maximum available power $P_{sub.available}$ is communicated to one or more other integrated PD modules **220p-q** by the module controller **402** via the digital communication bus Comm. Accordingly, at step **1004**, the amount of power $P_{sub.alloc.sup.p}$ allocated to port p is updated to $P_{sub.contract}$. (55) At step **1006**, if it is determined if a counter $Temp_{sup.p}$ of excess temperature events for port p has exceeded a first excess temperature event count threshold $T_{sub.warn}$, or that the power $P_{sub.alloc.sup.p}$ allocated to port p is already equal to a maximum amount of power $P_{sub.max.sup.p}$ that the integrated PD module at port p is able to deliver, flow of the process continues to step **1008**. USB temperature event detection is described in more detail below with reference to FIG. **1200**.

(56) At step **1008**, because the counter $Temp_{sup.p}$ of excess temperature events was greater than the first excess temperature event count threshold $T_{sub.warn}$, the USB Capability Mismatch flag for the sink device at port p is ignored by the integrated PD module associated with port p. Flow continues to step **1010**, where it is determined if the counter $Temp_{sup.p}$ of excess temperature events is greater than a second excess temperature event count threshold $T_{sub.critical}$. If the counter $Temp_{sup.p}$ of excess temperature events is greater than a second excess temperature event count threshold $T_{sub.critical}$, at step **1012**, USB-PD is disabled for port p. Flow then returns to step **506** shown in FIG. 5.

(57) If it was determined at step **1006** that the counter $Temp_{sup.p}$ of excess temperature events for port p had not exceeded the first excess temperature event count threshold $T_{sub.warn}$, and that the power $P_{sub.alloc.sup.p}$ allocated to port p was not already equal to the maximum amount of power $P_{sub.max.sup.p}$ that the integrated PD module at port p is able to deliver, flow of the process continues to step **1014**. At step **1014**, the USB capability mismatch field is copied by the associated integrated PD module (i.e., it is not ignored by the integrated PD module associated with port p). At step **1016**, a flag indicating that the PD contract negotiation at port p is complete is set at the associated integrated PD module. Flow additionally continues to step **1016** from step **1010**, described above, if it was determined at step **1010** that the counter $Temp_{sup.p}$ of excess temperature events is not greater than a second excess temperature event count threshold $T_{sub.critical}$. At step **1018**, capabilities for ports of the multi-port charger **201** other than port p are unmasked by the integrated PD modules of the multi-port charger **201**. Flow of the process then returns to step **506** shown in FIG. 5.

(58) FIG. **11** provides a portion of a simplified example process **1100** for adaptive power-sharing using the multi-port charger **201** shown in FIG. **2**, in accordance with some embodiments. The particular steps, the order of steps, and the combination of steps are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(59) Step **1102** of the process **1100** continues from step **516** shown in FIG. 5 and is performed in response to a determination at step **516** that a USB sink device disconnection has been detected at port p. At step **1102**, the amount of power $P_{sub.alloc.sup.p}$ that was previously allocated to port p is added back to the maximum available power $P_{sub.available}$ that remains to be distributed between the ports thereof. The adjustment in maximum available power $P_{sub.available}$ is communicated to one or more other integrated PD modules **220p-q** by the module controller **402**

via the digital communication bus Comm. At step **1104**, the amount of power $P_{sub.alloc.sup.p}$ allocated to port p is updated to 0 W (because no sink device is connected at port p). At step **1106**, a maximum current for port p is programmatically set to 1.5 A by configuring the programmatically controlled termination resistors **412** and signal multiplexing circuit **414**, using the PD controller **406** via the control signal CTRL.sup.4, to values indicative of R_p 1.5 (e.g., about 22 k Ohms), per the USB-PD standard if port p is configured as USB Type-C. At step **1108**, any status flags and counters that were associated with the sink device previously connected to port p, such as the counter Temp.sup.p of excess temperature events, are cleared by the integrated PD module associated with port p. At step **1110**, USB capability mismatch is unmasked, per the USB-PD standard, for ports other than port p.

(60) In some embodiments, at step **1112**, the integrated PD module associated with port p is advantageously operable to place the DC-to-DC power converter therein into a low-power consumption mode until a sink device is connected to port p. For example, the DC-to-DC power converter may be placed in a standby mode in which switching signals are disabled, thereby increasing an overall power efficiency of the multi-port charger **201** as compared to chargers that do enter a low-power mode. Flow of the process then returns to step **506** shown in FIG. 5.

(61) FIG. 12 provides a portion of a simplified example process **1200** for adaptive power-sharing using the multi-port charger **201** shown in FIG. 2, in accordance with some embodiments. The particular steps, the order of steps, and the combination of steps are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(62) The steps of process **1200** continue from step **506** shown in FIG. 5. At step **1202**, if an excess temperature warning event Temp.sub.warn was detected at port p by an associated integrated PD module, flow continues to step **1302**, shown in FIG. 13. Otherwise, flow continues to step **1204**. At step **1204**, if a normal temperature event Temp.sub.normal was detected at port p by an associated integrated PD module, flow continues to step **1402**, shown in FIG. 14. Otherwise, flow continues to step **1206**. At step **1206**, if a critical temperature event Temp.sub.critical was detected at port p by an associated integrated PD module, flow continues to step **1502**, shown in FIG. 15. Otherwise, flow continues to step **1208**. At step **1208**, if a sink capabilities event P.sub.sinkcap was detected at port p by an associated integrated PD module, flow continues to step **1602**, shown in FIG. 16. Otherwise, flow continues to step **1210**.

(63) At step **1210**, if it is determined, (e.g., using the controller modules thereof via the digital communication bus Comm), that a USB Less Power, No PD, flag is set for any port of the multi-port charger **201**, and that the maximum available power $P_{sub.available}$ that remains to be distributed between the ports thereof is greater than or equal to 7.5 W, flow returns to step **604** shown in FIG. 6. Otherwise, flow continues to step **1212**.

(64) At step **1212**, if it is determined, (e.g., using the controller modules thereof via the digital communication bus Comm), that a USB Less Power, No Quick Charge, flag is set for any port of the multi-port charger **201** and that the maximum available power $P_{sub.available}$ that remains to be distributed between the ports thereof is greater than or equal to 18 W minus the amount of power $P_{sub.alloc.sup.p}$ allocated to port p, flow returns to step **904** of FIG. 9. Otherwise, flow continues to step **1214**.

(65) At step **1214**, if it is determined, (e.g., using the controller modules thereof via the digital communication bus Comm), that a USB Capability Mismatch flag is set for port p and that capability mismatch is unmasked for any port of the multi-port charger **201**, flow continues to step **1702** shown in FIG. 17. Otherwise, flow returns to step **506** shown in FIG. 5.

(66) FIG. 13 provides a portion of a simplified example process **1300** for adaptive power-sharing using the multi-port charger **201** shown in FIG. 2, in accordance with some embodiments. The particular steps, the order of steps, and the combination of steps are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of

steps, and combinations of steps to achieve similar functions or results.

(67) Step **1302** of the process **1300** continues from step **1202** shown in FIG. **12** and is conducted in response to a determination that an excess temperature warning event Temp.sub.warn was detected at port p by an associated integrated PD module. At step **1302**, if it is determined that the counter Temp.sup.p of excess temperature events at port p is less than the first excess temperature event count threshold T.sub.warn, flow of the process continues to step **1304**. At step **1304**, the counter Temp.sup.p of excess temperature events at port p is incremented. At step **1306**, the maximum amount of power P.sub.max.sup.p that the integrated PD module at port p is permitted to deliver is reduced to a lower power level P.sub.lower.sup.p. At step **1308**, the maximum available power P.sub.available of the multi-port charger **201** that remains to be distributed between ports thereof is reduced by the new lower power level P.sub.lower.sup.p and the amount of power P.sub.alloc.sup.p previously allocated to port p is added back to the maximum available power P.sub.available. The adjustment in maximum available power P.sub.available is communicated to one or more other integrated PD modules **220p-q** by the module controller **402** via the digital communication bus Comm. Accordingly, at step **1310**, the amount of power P.sub.alloc.sup.p allocated to port p is updated to P.sub.lower.sup.p. At step **1312**, the USB-PD contract P.sub.contract.sup.p for port p is set to P.sub.alloc.sup.p. At step **1314**, USB-PD contract negotiation for port p is initiated by the integrated PD module associated with port p, in accordance with the USB-PD standard. At step **1316**, the integrated PD module associated with port p initiates a temperature re-check process for the sink device connected to port p. Flow of the process then returns to step **506** shown in FIG. **5**.

(68) If it was determined at step **1302** that the counter Temp.sup.p of excess temperature events at port p is not less than (i.e., is greater than or equal to) the first excess temperature event count threshold T.sub.warn, flow of the process continues to step **1318**. At step **1318**, all flags and counters associated with port p are cleared. At step **1320**, the maximum amount of power P.sub.max.sup.p that the integrated PD module at port p is permitted to deliver is set to 15 W. At step **1322**, the maximum available power P.sub.available of the multi-port charger **201** that remains to be distributed between the ports thereof is reduced by 15 W, and the amount of power P.sub.alloc.sup.p previously allocated to port p is added back to the maximum available power P.sub.available. The adjustment in maximum available power P.sub.available is communicated to one or more other integrated PD modules **220p-q** by the module controller **402** via the digital communication bus Comm. Accordingly, at step **1324**, the amount of power P.sub.alloc.sup.p allocated to port p is updated to 15 W. At step **1326**, a hard reset, in accordance with the USB-PD standard, is initiated by the integrated PD module associated with port p. Flow of the process then returns to step **506** shown in FIG. **5**.

(69) FIG. **14** provides a portion of a simplified example process **1400** for adaptive power-sharing using the multi-port charger **201** shown in FIG. **2**, in accordance with some embodiments. The particular step, the order in which the step is performed, and the combination of the step with other steps disclosed herein are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(70) Step **1402** of the process **1400** continues from step **1204** shown in FIG. **12** and is conducted in response to a determination that a normal temperature event Temp.sub.normal was detected at port p by an associated integrated PD module. Accordingly, at step **1402**, the counter Temp.sup.p of excess temperature events at port p is reset to 0. Flow of the process then returns to step **506** shown in FIG. **5**.

(71) FIG. **15** provides a portion of a simplified example process **1500** for adaptive power-sharing using the multi-port charger **201** shown in FIG. **2**, in accordance with some embodiments. The particular steps, the order of steps, and the combination of steps are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(72) Step **1502** of the process **1500** continues from step **1206** shown in FIG. **12** and is conducted in response to a determination that a critical excess temperature event Temp.sub.critical was detected at port p by an associated integrated PD module. At step **1502**, the counter Temp.sup.p of excess temperature events at port p is reset to 0. At step **1504**, a maximum current for port p is set to 3 A by configuring the programmatically controlled termination resistors **412** and signal multiplexing circuit **414**, using the PD controller **406** via the control signal CTRL.sup.4, to values indicative of R_p 3.0 (e.g., about 10 k Ohms), per the USB-PD standard. At step **1506**, the maximum available power P.sub.available of the multi-port charger **201** that remains to be distributed between the ports thereof is reduced by 15 W and the amount of power P.sub.alloc.sup.p previously allocated to port p is added back to the maximum available power P.sub.available. The adjustment in maximum available power P.sub.available is communicated to one or more other integrated PD modules **220p-q** by the module controller **402** via the digital communication bus Comm. Accordingly, at step **1508**, the amount of power P.sub.alloc.sup.p allocated to port p is updated to 15 W. At step **1510**, the USB-PD contract P.sub.contract.sup.p for port p is set to 15 W. At step **1512**, USB-PD contract negotiation for port p is initiated by the integrated PD module associated with port p, in accordance with the USB-PD standard. Flow of the process then returns to step **506** shown in FIG. **5**.

(73) FIG. **16** provides a portion of a simplified example process **1600** for adaptive power-sharing using the multi-port charger **201** shown in FIG. **2**, in accordance with some embodiments. The particular steps, the order of steps, and the combination of steps are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(74) Step **1602** of the process **1600** continues from step **1208** shown in FIG. **12** and is conducted in response to a P.sub.sinkcap power event being detected at port p by the associated integrated PD module. At step **1602**, if it is determined that the P.sub.sinkcap power is greater than the maximum available power P.sub.available of the multi-port charger **201** that remains to be distributed between the ports, plus the amount of power P.sub.alloc.sup.p previously allocated to port p, flow returns to step **506** shown in FIG. **5**. Otherwise, flow continues to step **1604**. At step **1604**, the maximum available power P.sub.available of the multi-port charger **201** that remains to be distributed between the ports thereof is reduced by the P.sub.sinkcap power, and the amount of power P.sub.alloc.sup.p previously allocated to port p is added back to the maximum available power P.sub.available. The adjustment in maximum available power P.sub.available is communicated to one or more other integrated PD modules **220p-q** by the module controller **402** via the digital communication bus Comm. Accordingly, at step **1606**, the amount of power P.sub.alloc.sup.p allocated to port p is updated to P.sub.sinkcap. At step **1608**, the USB-PD contract P.sub.contract.sup.p for port p is set to P.sub.alloc.sup.p. At step **1610**, USB-PD contract negotiation for port p is initiated by the integrated PD module associated with port p, in accordance with the USB-PD standard. At step **1612**, a Capability Mismatch flag for port p is cleared by the integrated PD module associated with port p. Flow of the process then returns to step **506** of FIG. **5**.

(75) FIG. **17** provides a portion of a simplified example process **1700** for adaptive power-sharing using the multi-port charger **201** shown in FIG. **2**, in accordance with some embodiments. The particular step, the order in which the step is performed, and the combination of the step with other steps disclosed herein are shown for illustrative and explanatory purposes only. Other embodiments can implement different particular steps, orders of steps, and combinations of steps to achieve similar functions or results.

(76) Step **1702** of the process **1700** continues from step **1214** shown in FIG. **12** and is conducted in response to determining by the integrated PD module associated with port p that a Capability Mismatch flag has been set for port p and that capability mismatch is unmasked for any port of the multi-port charger **201**. In response, at step **1702**, the integrated PD module associated with port p initiates a USB GetSinkCapabilities event for port p to receive P.sub.sinkcap power for the sink

device connected to port p, in accordance with the USB-PD standard.

(77) Reference has been made in detail to embodiments of the disclosed invention, one or more examples of which have been illustrated in the accompanying figures. Each example has been provided by way of explanation of the present technology, not as a limitation of the present technology. In fact, while the specification has been described in detail with respect to specific embodiments of the invention, it will be appreciated that those skilled in the art, upon attaining an understanding of the foregoing, may readily conceive of alterations to, variations of, and equivalents to these embodiments. For instance, features illustrated or described as part of one embodiment may be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present subject matter covers all such modifications and variations within the scope of the appended claims and their equivalents. These and other modifications and variations to the present invention may be practiced by those of ordinary skill in the art, without departing from the scope of the present invention, which is more particularly set forth in the appended claims. Furthermore, those of ordinary skill in the art will appreciate that the foregoing description is by way of example only, and is not intended to limit the invention.

Claims

1. A multi-port charger, comprising: a first power converter that generates a DC output voltage; and a plurality of integrated power delivery modules electrically coupled to the first power converter; wherein each of the integrated power delivery modules comprises: a module controller; a power delivery controller (“PD controller”) in signal communication with the module controller and which is configured to be connected to a sink device; a second power converter in signal communication with the module controller and the PD controller and which is configured to provide an output voltage to the sink device via a voltage bus; a first analog-to-digital converter (ADC) circuit in signal communication with the PD controller and the voltage bus to generate a digital representation of the output voltage provided by the second power converter to the sink device; and a second ADC circuit in signal communication with the PD controller and the voltage bus to provide a digital representation of an output current provided by the second power converter to the sink device.

2. The multi-port charger of claim 1, wherein: a first integrated power delivery module of the plurality of integrated power delivery modules is configured to determine a first target amount of power to be provided to a first sink device connected thereto by i) negotiating with the first sink device using a first PD controller of the first integrated power delivery module, and ii) negotiating with respective second module controllers of one or more second integrated power delivery modules of the plurality of integrated power delivery modules using a first module controller of the first integrated power delivery module.

3. The multi-port charger of claim 2, wherein: the first integrated power delivery module generates a digital representation of an actual amount of power delivered to the first sink device by the second power converter using the digital representation of the output voltage and the digital representation of an output current; and the first integrated power delivery module is configured to communicate the digital representation of the actual amount of power delivered to the first sink device to the second integrated power delivery modules using the first module controller to update an available amount of power that can be delivered by the multi-port charger.

4. The multi-port charger of claim 2, wherein: the first integrated power delivery module is configured to set the output voltage of the second power converter of the first integrated power delivery module in accordance with the first target amount of power via the first PD controller.

5. The multi-port charger of claim 4, wherein: upon determining, by the first PD controller, that the first sink device has been disconnected from the first integrated power delivery module, setting the second power converter of the first integrated power delivery module to a standby mode, the

standby mode comprising disabling switching signals of the second power converter.

6. The multi-port charger of claim 2, wherein: a first programmable configuration stored at one or more respective memory modules of the plurality of integrated power delivery modules specifies a total allowable power that may be delivered by the multi-port charger; a second integrated power delivery module of the one or more second integrated power delivery modules is configured to negotiate a second target amount of power to be provided to a second sink device connected thereto by i) negotiating with the second sink device using a second PD controller of the second integrated power delivery module, and ii) negotiating with the first integrated power delivery module using a second module controller of the second integrated power delivery module; and a summation of the first target amount of power and the second target amount of power is less than or equal to the total allowable power.

7. The multi-port charger of claim 6, wherein: upon determining by the second integrated power delivery module, from the first PD controller, that a third target amount of power should be provided to the first sink device: the second integrated power delivery module is configured to negotiate a fourth target amount of power to be provided to the second sink device connected thereto by i) negotiating with the second sink device using the second PD controller of the second integrated power delivery module, and ii) negotiating with the first integrated power delivery module using the second module controller of the second integrated power delivery module; and a summation of the third target amount of power and the fourth target amount of power is less than or equal to the total allowable power.

8. The multi-port charger of claim 7, wherein: each integrated power delivery module of the plurality of integrated power delivery modules comprises a respective third ADC circuit in signal communication with a respective PD controller of that integrated power delivery module and a respective temperature sensing circuit to generate a digital representation of a temperature value of that integrated power delivery module; and the third target amount of power is based on the digital representation of the temperature value.

9. The multi-port charger of claim 7, wherein: the third target amount of power is based on a charge level of a battery of the first sink device.

10. The multi-port charger of claim 7, wherein: the third target amount of power is based on a priority of the first sink device as compared to the second sink device.

11. An integrated power delivery module, comprising: a module controller; a power delivery controller ("PD controller") in signal communication with the module controller and which is configured to be connected to a sink device; a power converter in signal communication with the module controller and the PD controller and which is configured to receive an input voltage and to provide an output voltage to the sink device via a voltage bus; a first analog-to-digital converter (ADC) circuit in signal communication with the PD controller and the voltage bus to generate a digital representation of the output voltage provided by the power converter to the sink device; and a second ADC circuit in signal communication with the PD controller and the voltage bus to provide a digital representation of an output current provided by the power converter to the sink device.

12. The integrated power delivery module of claim 11, wherein: a first integrated power delivery module of a plurality of integrated power delivery modules is configured to determine a first target amount of power to be provided to a first sink device connected thereto by i) negotiating with the first sink device using a first PD controller of the first integrated power delivery module, and ii) negotiating with respective second module controllers of one or more second integrated power delivery modules of the plurality of integrated power delivery modules using a first module controller of the first integrated power delivery module.

13. The integrated power delivery module of claim 12, wherein: the first integrated power delivery module is configured to generate a digital representation of an actual amount of power delivered to the first sink device by the power converter using the digital representation of the output voltage

and the digital representation of an output current; and the first integrated power delivery module is configured to communicate the digital representation of the actual amount of power delivered to the first sink device to the second integrated power delivery modules using the first module controller to update an available amount of power that can be delivered by the plurality of integrated power delivery modules.

14. The integrated power delivery module of claim 12, wherein: the first integrated power delivery module is configured to set the output voltage of a first power converter of the first integrated power delivery module in accordance with the first target amount of power via the first PD controller.

15. The integrated power delivery module of claim 14, wherein: upon determining, by the first PD controller, that the first sink device has been disconnected from the first integrated power delivery module, setting the first power converter to a standby mode, the standby mode comprising disabling switching signals of the first power converter.

16. The integrated power delivery module of claim 12, wherein: a first programmable configuration stored at one or more respective memory modules of the plurality of integrated power delivery modules specifies a total allowable power that may be delivered by the plurality of integrated power delivery modules; a second integrated power delivery module of the one or more second integrated power delivery modules is configured to negotiate a second target amount of power to be provided to a second sink device connected thereto by i) negotiating with the second sink device using a second PD controller of the second integrated power delivery module, and ii) negotiating with the first integrated power delivery module using a second module controller of the second integrated power delivery module; and a summation of the first target amount of power and the second target amount of power is less than or equal to the total allowable power.

17. The integrated power delivery module of claim 16, wherein: upon determining by the second integrated power delivery module, from the first PD controller, that a third target amount of power should be provided to the first sink device: the second integrated power delivery module is configured to negotiate a fourth target amount of power to be provided to the second sink device connected thereto by i) negotiating with the second sink device using the second PD controller of the second integrated power delivery module, and ii) negotiating with the first integrated power delivery module using the second module controller of the second integrated power delivery module; and a summation of the third target amount of power and the fourth target amount of power is less than or equal to the total allowable power.

18. The integrated power delivery module of claim 17, wherein: each integrated power delivery module of the plurality of integrated power delivery modules comprises a respective third ADC circuit in signal communication with a respective PD controller of that integrated power delivery module and a respective temperature sensing circuit to generate a digital representation of a temperature value of that integrated power delivery module; and the third target amount of power is based on the digital representation of the temperature value.

19. The integrated power delivery module of claim 17, wherein: the third target amount of power is based on a charge level of a battery of the first sink device.

20. The integrated power delivery module of claim 17, wherein: the third target amount of power is based on a priority of the first sink device as compared to the second sink device.
